

A MACHINE-CHECKED CONSTRUCTION OF THE UHF ALGEBRA M_{3^∞} WITH A FAITHFUL TRACIAL STATE: FORMALIZING GLIMM SIMPLICITY IN LEAN 4

PABLO COHEN

ABSTRACT. We report a complete Lean 4 formalization, over mathlib [8, 9], of the uniformly hyperfinite (UHF) C^* -algebra M_{3^∞} — realized as the metric completion T_∞ of the inductive limit of the matrix tower $M_{3^k}(\mathbb{C})$ under the unital $*$ -embeddings $A \mapsto A \otimes I_3$ — together with a canonical tracial state τ_{UHF} on T_∞ that is kernel-verified to be additive, unital, 1-Lipschitz, tracial, Hermitian, positive, and *faithful*: $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ for every $x \in T_\infty$. As a corollary we obtain a machine-checked proof that T_∞ is algebraically simple — every two-sided ideal is $\perp$ or $\top$. We further prove that τ_{UHF} is the *unique* continuous unital tracial functional on T_∞ (theorem `substrate_UHF_trace_unique`; positivity is not required). Faithfulness, simplicity, and uniqueness of the trace together identify T_∞ as the Glimm 3^∞ UHF factor. To the best of our knowledge, after searching the major proof-assistant libraries (mathlib, the Coq `opam` archive, and the Isabelle Archive of Formal Proofs) and the formalization literature, this is the first formalization of Glimm’s 1960 simplicity theorem for a UHF algebra [6] and the first machine-checked construction of an infinite-dimensional simple C^* -algebra with a canonical faithful and unique trace; we have not performed an exhaustive survey and would welcome pointers to prior art.

The completion-tier faithfulness proof is the heart of the paper. It routes through a trace-preserving conditional expectation $E_k: T_\infty \rightarrow M_{3^k}(\mathbb{C})$ (the partial trace over the tail), proved to be a $\|\cdot\|_2$ -contraction by an elementary Cauchy–Schwarz argument — where the classical route would invoke the Kadison–Schwarz inequality and complete positivity, neither of which is available in mathlib — and an operator-norm contraction via the isometry decomposition $E_k B = \frac{1}{3} \sum_t W_t^* B W_t$. Supporting machinery, novel as formalized content, includes a Cauchy–Schwarz inequality for the trace pairing, the trace-null set as a mathlib-native two-sided ideal, a discrete Weyl clock–shift unitary-averaging engine $\frac{1}{n^2} \sum_{a,b} (C^a S^b) x (C^a S^b)^* = (\text{tr } x/n) \cdot 1$, algebraic simplicity of the pre-completion tower, and a four-fold equivalence (trace-detection $\Leftrightarrow$ triviality of the trace-null ideal $\Leftrightarrow$ faithfulness $\Leftrightarrow$ simplicity) that precisely located the difficulty before it was overcome. Three standalone API files of mathlib-PR quality (topological closures of two-sided ideals; the non-unital continuous functional calculus mapping into closed ideals; spectral projections from clopen quasispectral sets) are contributed as by-products. Every headline theorem reports kernel axioms [`propext`, `Classical.choice`, `Quot.sound`], with zero project axioms and zero `sorry`s, verified at pinned toolchain and mathlib revisions.

CONTENTS

1. Introduction	2
1.1. Contributions	3
1.2. Why faithfulness is the hard part	3
1.3. An honest development history	3
1.4. A note on naming	4
1.5. Outline	4
2. Mathematical background	4
2.1. UHF algebras	4
2.2. Tracial states	5
2.3. Simplicity	5
2.4. Conditional expectations	5
3. The tower, the limit, and the completion	5
3.1. The tower	5

Date: July 23, 2026.

3.2. The inductive limit and its norm	5
3.3. The completion	6
4. The tracial state and its soft properties	6
4.1. Construction	6
4.2. The soft properties	6
4.3. The three tiers of faithfulness	7
5. The simplicity/faithfulness engine	7
5.1. Cauchy–Schwarz for the trace pairing (r102)	7
5.2. The trace-null two-sided ideal (r103)	7
5.3. The Weyl unitary-averaging engine (r104)	8
5.4. Algebraic simplicity below the completion (r105)	8
5.5. Averaging inside the completion (r106)	8
5.6. The four-fold equivalence (r108): locating the mountain	9
6. The conditional expectation and the faithfulness proof	9
6.1. The partial trace E_k (r109)	9
6.2. The $\ \cdot\ _2$ -contraction, elementarily (r110)	9
6.3. The operator-norm contraction via isometries (r111)	10
6.4. Extension to the completion and the closing argument (r112)	10
6.5. Uniqueness of the trace (r113)	11
7. Formalization notes	12
7.1. Mathlib infrastructure used	12
7.2. Gaps found in mathlib	12
7.3. Contributed API: the <code>ForMathlib</code> files	13
7.4. Companion result	13
8. Reproducibility	13
9. Related and future work	14
9.1. Related work	14
9.2. Future work	14
Acknowledgments	15
References	15

1. INTRODUCTION

Uniformly hyperfinite (UHF) C^* -algebras are among the oldest and most completely understood objects of operator algebra theory. Introduced by Glimm in 1960 [6] — with roots in the hyperfinite factor constructions of Murray and von Neumann [10] — they are the norm-closures of increasing unions of full matrix algebras, classified completely by a supernatural number, each simple, each carrying a unique tracial state which is automatically faithful. Standard references establish these facts [4, 1, 13]. None of them, before the present work, had been machine-checked.

The gap is not accidental. Formalized operator algebra is young: `mathlib` — the Lean 4 mathematical library [8, 9] — provides a C^* -algebra typeclass hierarchy, an L^2 -operator-norm C^* -structure on matrix algebras, ring direct limits, metric completions, and a continuous functional calculus, largely following the formalized functional analysis program of Dupuis, Lewis, and Macbeth [5]. But it contains no concrete infinite-dimensional simple C^* -algebra, no tracial state theory, no conditional expectations, no completely positive maps, and no spectral projections. Constructing a UHF algebra and proving the classical facts about it therefore requires building, from `mathlib`’s primitives, a substantial amount of infrastructure that the pen-and-paper literature takes for granted — and, at several points, finding proofs *different from* the classical ones, because the classical proofs lean on theory (Kadison–Schwarz, complete positivity, Borel functional calculus) that does not yet exist in any proof assistant library.

1.1. **Contributions.** This paper describes a Lean 4 formalization, kernel-verified with axioms `[propext, Classical.choice, Quot.sound]` only, zero project-level axioms, and zero `sorry`s, of:

- (C1) **The UHF algebra M_{3^∞} , `mathlib-native`.** The base-3 matrix tower $A_k := M_{3^k}(\mathbb{C})$ with the unital $*$ -embeddings $A \mapsto \text{reindex}(A \otimes I_3)$, its ring direct limit, and the metric completion T_∞ of that limit under the L^2 -operator norm, carrying `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances (§3).
- (C2) **A canonical tracial state τ_{UHF} on T_∞ ,** obtained as the unique uniformly continuous extension of the normalised matrix traces, and kernel-verified to be additive, unital, 1-Lipschitz, tracial, Hermitian, and positive (§4).
- (C3) **Unconditional faithfulness of τ_{UHF} at every tier** — matrix level, direct limit, and, the hard case, the completion: $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ for all $x \in T_\infty$ (theorem `UHF_trace_faithful`, §6).
- (C4) **Glimm simplicity of T_∞ :** every two-sided ideal of T_∞ is $\perp$ or $\top$ (theorem `substrate_completion_simple_unconditional`, §6). To the best of our knowledge — after searching the major proof-assistant libraries and literature (see the priority note in §9) — this is the first formalized simplicity theorem for an infinite-dimensional C^* -algebra.
- (C5) **Uniqueness of the trace:** τ_{UHF} is the unique continuous unital tracial functional on T_∞ (theorem `substrate_UHF_trace_unique`, §6); the argument requires only linearity, the trace identity, unitality, and continuity, not positivity. Together with faithfulness and simplicity this identifies T_∞ as the Glimm 3^∞ UHF factor (`r113_substrate_UHF_factor_capstone`).
- (C6) **The proof machinery itself,** novel as formalized content: a Cauchy–Schwarz inequality for the trace sesquilinear pairing; the trace-null set as a `mathlib-native` `TwoSidedIdeal`; a discrete Weyl clock–shift unitary-averaging engine; algebraic simplicity of the pre-completion tower; a completion-tier averaging operator; a four-fold equivalence between trace-detection, ideal-triviality, faithfulness, and simplicity; and a trace-preserving conditional expectation $E_k: T_\infty \rightarrow M_{3^k}(\mathbb{C})$ with elementary $\|\cdot\|_2$ - and operator-norm contraction proofs (§§5–6). Three standalone `mathlib`-PR-quality API files are contributed as by-products (§7).

1.2. **Why faithfulness is the hard part.** Everything about τ_{UHF} *except* faithfulness transfers from the matrix level to the completion by soft means. Additivity, traciality, Hermitian symmetry, and positivity are all *closed conditions*: each asserts that a continuous function vanishes (or is nonnegative) on T_∞ , so it suffices to verify it on the dense image of the direct limit and invoke completion induction on a closed set. Matrix-level verification is elementary, and `mathlib`’s `UniformSpace.Completion.induction_on` closes each one.

Faithfulness is different in kind. The statement $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ is *not* a closed condition — the set of x satisfying an implication with a nontrivial conclusion is not the preimage of a closed set — so density arguments do not apply. A limit of matrices, each individually detected by its trace, could in principle converge to a nonzero element whose trace-of-square vanishes: ruling this out requires genuinely global structure. Classically one appeals to either (a) simplicity of the algebra — a state whose left kernel generates a proper closed two-sided ideal must be faithful when no such ideal exists — with simplicity supplied by Glimm’s theorem [6]; or (b) the norm-one conditional expectations onto the finite levels, in the tradition of Tomiyama [14], whose contractivity is classically a consequence of complete positivity and the Kadison–Schwarz inequality [7]. Neither route existed in `mathlib`. The formalization ultimately built route (b) from scratch, with elementary replacements for the missing complete-positivity theory (§6), and obtained route (a)’s conclusion — simplicity — as a corollary rather than an input, inverting the classical dependency.

1.3. **An honest development history.** We describe the development arc candidly, because we believe the trajectory is itself a datum about formalizing operator algebras, and because

an earlier version of this material contained a claim we had to retract before it was eventually proved.

The construction of T_∞ and of τ_{UHF} with its six soft properties was completed in early July 2026. At that stage the source repository also contained a completion-tier “faithfulness” declaration that, on inspection, was a placeholder: a named proposition of type `Prop := True` discharged by `trivial`, together with a theorem whose hypothesis was completion-tier faithfulness restated. Neither constitutes a proof of anything, and a revision of the companion ITP note (2026-07-20) explicitly demoted completion-tier faithfulness from “conditional” to *open*, documenting the placeholder for transparency.

The subsequent closing arc, labelled r102–r112 in the repository, then unfolded in three phases:

- (1) *Reduction* (r102–r106): Cauchy–Schwarz for the trace pairing, the trace-null two-sided ideal, the Weyl averaging engine, algebraic simplicity of the pre-completion tower, and a completion-tier averaging operator together reduced faithfulness, in kernel-verified form, to a single named hypothesis: *trace-detection of ideals* (every nonzero two-sided ideal contains an element of nonzero trace).
- (2) *Obstruction analysis* (r108): a kernel-verified four-fold equivalence showed that trace-detection is not weaker than the goal — it is *logically equivalent* to faithfulness, and to simplicity. The reduction had restated the mountain, not descended it. This negative result was formalized rather than discarded: it is the precise statement of where the classical hard input lives.
- (3) *Closure* (r109–r112): the trace-preserving conditional expectation E_k (the tail partial trace) was constructed on the direct limit, proved a $\|\cdot\|_2$ -contraction by elementary Cauchy–Schwarz (where Kadison–Schwarz was expected to be needed), proved an operator-norm contraction via an isometry decomposition (where complete positivity was expected to be needed), extended to the completion, and used to close faithfulness unconditionally — with simplicity following through the r108 equivalence.

The placeholder was thus first exposed, then reduced, then genuinely closed. We report this sequence as a strength of the machine-checked setting: at every stage the kernel-verified statement of *what had actually been proved* was unambiguous, the gap between claim and theorem was mechanically visible, and the final theorems are immune to the wishful reading that produced the original placeholder.

1.4. A note on naming. The Lean development is part of a larger source project, and some identifiers inherit that project’s naming scheme: the direct limit is called `TimelessFieldRing` and its completion `TimelessFieldCompletion` in the source. These are names, nothing more; in this paper the objects are denoted $\varinjlim_k A_k$ and T_∞ , and every property claimed of them is one of the kernel-verified theorems listed below or a standard cited fact. Nothing in this paper depends on any framework, interpretation, or conjecture of the broader source project.

1.5. Outline. §2 recalls the classical background. §3 constructs the tower, limit, and completion. §4 constructs τ_{UHF} and its soft properties. §5 develops the simplicity/fidelity engine: Cauchy–Schwarz, the trace-null ideal, Weyl averaging, pre-completion simplicity, and the four-fold equivalence. §6 is the heart: the conditional expectation and the fidelity proof. §7 records formalization notes: mathlib APIs used, gaps found, and the three contributed API files. §8 gives the pinned toolchain and verification data. §9 discusses related and future work.

2. MATHEMATICAL BACKGROUND

This section fixes classical terminology for the formalization audience; all results here are standard [6, 4, 1, 13] and are *not* claimed as contributions. Which of them are formalized, and which are merely cited, is stated precisely in §§3–6.

2.1. UHF algebras. A *uniformly hyperfinite (UHF) algebra* is a unital C^* -algebra A containing an increasing sequence of unital $*$ -subalgebras $A_0 \subseteq A_1 \subseteq \cdots$, each $*$ -isomorphic to a full matrix algebra $M_{n_k}(\mathbb{C})$, whose union is dense in A . Necessarily $n_k \mid n_{k+1}$. Glimm [6] classified UHF

algebras by the associated supernatural number and proved each is simple with a unique tracial state; Bratteli [2] extended the framework to general approximately finite-dimensional (AF) algebras. The algebra studied here is M_{3^∞} : the case $n_k = 3^k$ with the embeddings $A \mapsto A \otimes I_3$. The choice of base 3 over the more common CAR tower M_{2^∞} is inherited from the source project's surrounding ternary constructions; as a classification target the two are equally standard, and every argument below works verbatim in any base ≥ 2 .

2.2. Tracial states. A *tracial state* on a unital C^* -algebra A is a linear functional $\tau: A \rightarrow \mathbb{C}$ with $\tau(1) = 1$, $\tau(x^*x) \geq 0$, and $\tau(xy) = \tau(yx)$. It is *faithful* if $\tau(x^*x) = 0$ implies $x = 0$. On $M_n(\mathbb{C})$ the unique tracial state is the normalised trace $\tau_n = \text{tr}/n$, and its faithfulness is the statement that $\text{tr}(M^*M) = \sum_{i,j} |M_{ij}|^2$ vanishes only at $M = 0$. On a UHF algebra the normalised traces are compatible with the embeddings and extend to the unique tracial state on the completion; its faithfulness is classically deduced from simplicity (the closure of the left kernel is a proper closed two-sided ideal) or from the conditional expectations onto the finite levels.

2.3. Simplicity. A ring is (algebraically) *simple* if its only two-sided ideals are 0 and the whole ring. For C^* -algebras one often restricts attention to closed ideals, but UHF algebras satisfy the stronger algebraic statement, and it is the algebraic statement that is formalized here: *every* two-sided ideal of T_∞ , closed or not, is $\perp$ or $\top$. The engine of the classical proof is finite-dimensional: averaging over a suitable finite group of unitaries in $M_n(\mathbb{C})$ projects any matrix onto its normalised trace times the identity, and since two-sided ideals absorb conjugation, sums, and scalars, a nonzero ideal element with nonzero trace forces 1 into the ideal. The finite unitary family used here is the discrete Weyl (clock-and-shift) pair [15]: $C = \text{diag}(1, \omega, \dots, \omega^{n-1})$ with $\omega = e^{2\pi i/n}$, and S the cyclic shift.

2.4. Conditional expectations. A *conditional expectation* of a C^* -algebra A onto a subalgebra B is a positive, unital, B -bimodule projection $E: A \rightarrow B$; by Tomiyama's theorem [14] these are exactly the norm-one projections. For a UHF algebra, the canonical conditional expectation onto the level- k subalgebra is the *partial trace over the tail*: under $M_{3^{k+1}} \cong M_{3^k} \otimes M_3$, it is $\text{id} \otimes \tau_3$. Classically, a conditional expectation is completely positive, hence satisfies the Kadison–Schwarz inequality $E(x)^*E(x) \leq E(x^*x)$ [7], from which its $\|\cdot\|_2$ -contractivity with respect to a compatible trace follows, and its complete positivity together with unitality gives $\|E\| = 1$ (a fact often organised via the Stinespring representation [12]). The formalization proves the two contraction properties it needs *directly*, without any of this theory; see §6.2 and §6.3.

3. THE TOWER, THE LIMIT, AND THE COMPLETION

3.1. The tower. At level $k \in \mathbb{N}$, define

$$A_k := \text{Matrix}(\text{Fin } 3^k)(\text{Fin } 3^k) \mathbb{C},$$

a finite-dimensional C^* -algebra under the L^2 -operator norm (mathlib's `Matrix.Norms.L2Operator` scoped instance, which provides the C^* -identity $\|M^*M\| = \|M\|^2$). Successive levels embed by the ternary tensor construction

$$\iota_k: A_k \hookrightarrow A_{k+1}, \quad A \mapsto \text{reindex}(\text{levelStepEquiv } k)(A \otimes I_3),$$

where `levelStepEquiv` k is the equivalence $\text{Fin } 3^k \times \text{Fin } 3 \simeq \text{Fin } 3^{k+1}$ induced by the ternary decomposition of 3^{k+1} , and $\otimes$ is the Kronecker product. In the Lean source this map is `substrateEmbedMatrix`, with iterated form `substrateRingHomIter` $m \rightarrow k$ for $m \leq k$. Each ι_k is a unital $*$ -ring homomorphism and an isometry.

3.2. The inductive limit and its norm. Mathlib's `Ring.DirectLimit` applied to $\{A_k, \iota_k\}$ produces the $\mathbb{C}$ -algebra

$$R := \varinjlim_k A_k \quad (\text{Lean: } \text{TimelessFieldRing}),$$

with `StarRing` structure descending from entrywise conjugate transpose. Every element of R has a single-level representative: mathlib's `DirectLimit.exists_eq_mk` produces, from $x \in R$,

a level k and a matrix $A \in A_k$ with $x = \llbracket(k, A)\rrbracket$. Because the embeddings are isometric, the level norm is well-defined on R (`substrateLevelToTimelessField_opNorm_eq`) and makes R a normed $*$ -ring; the canonical level embedding $\iota_k: A_k \rightarrow R$ (Lean: `substrateLevelToTimelessField`) is an isometry (theorem `substrateLevelToTimelessField_isometry`), and the cocone identity $\iota_k \circ (\text{iterated embedding } m \rightarrow k) = \iota_m$ holds definitionally up to `Quotient.sound` (`substrateLevelToTimelessField_iter`).

3.3. The completion. The metric completion of R under the operator norm is

$$T_\infty := \overline{\lim_k A_k}^{\|\cdot\|_{\text{op}}} \quad (\text{Lean: TimelessFieldCompletion}),$$

built with `UniformSpace.Completion`. The algebraic operations extend by uniform continuity, and the C^* -identity passes to the completion by the classical Cauchy-sequence argument, executed against `mathlib`'s completion API.

Theorem 3.1 (UHF C^* -algebra registration). *T_∞ carries `mathlib-native` `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances, with $\varinjlim_k A_k$ dense in T_∞ . Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

By construction T_∞ is a UHF algebra in the classical sense: the norm-closure of an increasing union of full matrix algebras under unital $*$ -embeddings, of supernatural number 3^∞ . We emphasise that the `mathlib CStarAlgebra` instance is the algebra of record; no appeal is made to the classical classification.

Theorem 3.2 (Level-wise simplicity). *For every k , $A_k = M_{3^k}(\mathbb{C})$ is a `mathlib IsSimpleRing` instance, via `DivisionRing.isSimpleRing` on $\mathbb{C}$ composed with `IsSimpleRing.matrix`.*

4. THE TRACIAL STATE AND ITS SOFT PROPERTIES

4.1. Construction. On each level the normalised trace

$$\tau_n: A_k \rightarrow \mathbb{C}, \quad \tau_n(M) := \frac{\text{trace}(M)}{n}, \quad n = 3^k$$

is linear, unital, and 1-Lipschitz for the operator norm (via the Hilbert–Schmidt bound $\|M\|_{\text{HS}}^2 \leq n\|M\|_{\text{op}}^2$ and Cauchy–Schwarz). Compatibility with the embeddings, $\tau_{3^{k+1}}(\iota_k A) = \tau_{3^k}(A)$, follows from `Matrix.trace_kronecker`. The universal property of the direct limit (`DirectLimit.lift`) produces the pre-trace $\tau_{\text{pre}}: R \rightarrow \mathbb{C}$ (Lean: `substrate_pre_trace`), additive, unital, and 1-Lipschitz, hence uniformly continuous; and

$$\tau_{\text{UHF}} := \text{UniformSpace.Completion.extension}(\tau_{\text{pre}}): T_\infty \rightarrow \mathbb{C} \quad (\text{Lean: UHF_trace})$$

is its unique continuous extension, agreeing with τ_{pre} on the dense image (`UHF_trace_coe`).

4.2. The soft properties.

Theorem 4.1 (The canonical tracial state). *$\tau_{\text{UHF}}: T_\infty \rightarrow \mathbb{C}$ satisfies, kernel-verified:*

- (i) additive, unital, 1-Lipschitz, uniformly continuous;
- (ii) $\mathbb{C}$ -linear: $\tau_{\text{UHF}}(c \cdot x) = c \tau_{\text{UHF}}(x)$ (`UHF_trace_smul`; bundled as `UHF_trace_linearMap` : $T_\infty \rightarrow_{\mathbb{C}} \mathbb{C}$);
- (iii) tracial: $\tau_{\text{UHF}}(xy) = \tau_{\text{UHF}}(yx)$ (`UHF_trace_mul_comm`);
- (iv) Hermitian: $\tau_{\text{UHF}}(x^*) = \overline{\tau_{\text{UHF}}(x)}$;
- (v) positive: $0 \leq \tau_{\text{UHF}}(x^*x)$ in the complex order (`UHF_trace_star_mul_self_nonneg`, under `mathlib`'s `scoped ComplexOrder`).

Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.

Each of (i)–(v) is a closed condition and is proved by the same three-step pattern: a `mathlib` matrix lemma at level k (`Matrix.trace_mul_comm`, `Matrix.trace_conjTranspose`, `Matrix.posSemidef_conjTranspose_mul_self` with `Matrix.PosSemidef.trace_nonneg`); reduction of direct-limit elements to single-level representatives via `DirectLimit.exists_eq_mk`; and completion induction (`UniformSpace.Completion.induction_on`, or its binary form for (iii))

on the closed set where the identity holds — closedness of the positivity set (v) uses `Complex.orderClosedTopology`.

4.3. The three tiers of faithfulness. Faithfulness stratifies over the construction:

- (Φ1) *Matrix level.* $\tau_n(M^*M) = 0 \Rightarrow M = 0$, from `mathlib's Matrix.trace_conjTranspose_mul_self_eq_zero_iff` (Lean: `normalized_matrix_trace_star_mul_self_eq_zero_iff`). Elementary: the trace of M^*M is the sum of $|M_{ij}|^2$.
- (Φ2) *Direct limit.* $\tau_{\text{pre}}(x^*x) = 0 \Rightarrow x = 0$ (`substrate_pre_trace_star_mul_self_faithful`): reduce x to a single-level representative, apply (Φ1), and push 0 back through the embedding.
- (Φ3) *Completion.* $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ for $x \in T_\infty$. **This is the hard tier**, for the reason explained in §1.2: it is not a closed condition, and no density argument applies. Its unconditional proof is the subject of §§5–6.

5. THE SIMPLICITY/FAITHFULNESS ENGINE

This section develops the machinery that first *reduced* completion-tier faithfulness to a single named hypothesis, and then *identified* that hypothesis as equivalent to the goal. All results here are unconditional kernel-verified theorems; the r-labels record the repository development arc but the content is mathematics, not changelog.

5.1. Cauchy–Schwarz for the trace pairing (r102). Define the trace sesquilinear pairing $\langle x, y \rangle_\tau := \tau_{\text{UHF}}(y^*x)$ on T_∞ (Lean: `traceInner`). Positivity (Theorem 4.1(v)) makes it a positive semi-definite sesquilinear form, and the classical discriminant argument goes through against `mathlib's discriminant_le_zero`:

Theorem 5.1 (`UHF_trace_cauchySchwarz`). *For all $x, y \in T_\infty$,*

$$\|\tau_{\text{UHF}}(y^*x)\|^2 \leq \text{Re } \tau_{\text{UHF}}(x^*x) \cdot \text{Re } \tau_{\text{UHF}}(y^*y).$$

The proof expands $\tau_{\text{UHF}}((x - cy)^*(x - cy)) \geq 0$ along real multiples $c = t \tau_{\text{UHF}}(y^*x)$ using completion-tier linearity (Theorem 4.1(ii)) and the `StarModule` structure, and applies the quadratic discriminant bound. An immediate corollary is left absorption of the trace-null cone (`UHF_trace_null_left_absorption`): $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow \tau_{\text{UHF}}((ax)^*(ax)) = 0$ for every a , since $\tau_{\text{UHF}}((ax)^*(ax)) = \langle a^*ax, x \rangle_\tau$ is dominated through Theorem 5.1 by $\text{Re } \tau_{\text{UHF}}(x^*x) = 0$.

5.2. The trace-null two-sided ideal (r103). Let $N := \{x \in T_\infty : \tau_{\text{UHF}}(x^*x) = 0\}$. Theorem 5.1 makes N additive (the cross terms in the sesquilinear expansion of $\tau_{\text{UHF}}((x + y)^*(x + y))$ are killed by Cauchy–Schwarz), traciality makes it $*$ -closed ($\tau_{\text{UHF}}(xx^*) = \tau_{\text{UHF}}(x^*x)$), left absorption plus $*$ -closure gives right absorption, and scalar closure is immediate. The result is bundled as a `mathlib-native` object:

Theorem 5.2 (`traceNullTwoSidedIdeal`). *N is the carrier of a `TwoSidedIdeal` of T_∞ , and it is proper: $1 \notin N$ since $\tau_{\text{UHF}}(1) = 1$ (`traceNullTwoSidedIdeal_ne_top`).*

This immediately yields the conditional form of faithfulness — the honest theorem available at that stage of the development:

Theorem 5.3 (`UHF_trace_faithful_of_simple`). *If every two-sided ideal of T_∞ is $\perp$ or $\top$, then τ_{UHF} is faithful.*

Proof sketch. N is a proper two-sided ideal, so simplicity forces $N = \perp$; and $x \in N$ means exactly $\tau_{\text{UHF}}(x^*x) = 0$, so membership in $\perp$ gives $x = 0$. The hypothesis is genuine algebraic simplicity of T_∞ — classically Glimm’s theorem — not a self-referential marker. $\square$

5.3. The Weyl unitary-averaging engine (r104). The engine of every simplicity proof for matrix towers is finite-dimensional linear algebra. Fix $n \geq 1$ and let $\omega := e^{2\pi i/n}$ (Lean: `subtrateRoot`, a primitive n -th root by `Complex.isPrimitiveRoot_exp`). Define the *clock* and *shift* matrices [15]

$$C := \text{diagonal}(j \mapsto \omega^j), \quad S_{ij} := [i = j + 1],$$

both unitary (`clockMatrix_mem_unitaryGroup`, `shiftMatrix_mem_unitaryGroup`), with closed-form powers $C^m = \text{diagonal}(j \mapsto \omega^{jm})$ and $(S^m)_{ij} = [i = j + m]$. The character orthogonality relation $\sum_{a < n} (\omega^i \overline{\omega^j})^a = n [i = j]$ (`substrate_clock_character_sum`, via `geom_sum_mul` and `IsPrimitiveRoot.pow_inj`) drives a two-stage computation:

Lemma 5.4 (Stage A: `clock_average_eq_diagonalPart`). *Conjugation-averaging over the clock powers kills the off-diagonal: $\frac{1}{n} \sum_{a < n} C^a x (C^a)^* = \text{diagonal}(\text{diag } x)$.*

Lemma 5.5 (Stage B: `shift_average_diagonal_eq_trace_smul_one`). *Conjugation-averaging over the shift powers equalises the diagonal: for diagonal D , $\frac{1}{n} \sum_{b < n} S^b D (S^b)^* = (\text{trace } D/n) \cdot 1$.*

Theorem 5.6 (The engine: `matrix_unitary_average_eq_trace_smul_one`). *For every $x \in M_n(\mathbb{C})$,*

$$\frac{1}{n^2} \sum_{a < n} \sum_{b < n} (C^a S^b) x (C^a S^b)^* = \frac{\text{trace } x}{n} \cdot 1.$$

Since a set closed under addition, arbitrary left/right multiplication, and $\mathbb{C}$ -scaling absorbs each conjugate uxu^* , it absorbs the average (`trace_smul_one_mem_of_two_sided_absorbing`): any two-sided absorbing set containing x contains $(\text{trace } x/n) \cdot 1$.

5.4. Algebraic simplicity below the completion (r105). Firing the engine at a nonzero element requires a nonzero *trace*, which a nonzero ideal element need not have. The fix is the classical matrix-unit trick: if $x \neq 0$ pick a nonzero entry x_{pq} ; the matrix unit $E_{q,p}$ satisfies $\text{trace}(E_{q,p} x) = x_{pq} \neq 0$ (`trace_single_mul`), and $E_{q,p} x$ is still in the absorbing set. Averaging and rescaling puts 1 in the set (`absorbing_set_contains_one`). Together with Theorem 3.2 this re-derives matrix-level simplicity in ideal form (`matrix_twoSidedIdeal_bot_or_top`) and, by pulling a two-sided ideal $I \leq R$ back to each level (`substrateIdealPullback`, with the four closure lemmas), lifts it to the direct limit:

Theorem 5.7 (`directLimit_twoSidedIdeal_bot_or_top`). *Every two-sided ideal of $\varinjlim_k M_{3^k}(\mathbb{C})$ is $\perp$ or $\top$.*

5.5. Averaging inside the completion (r106). The Weyl words at level k push forward to unitaries $u_{a,b} \in T_\infty$, and the completion-tier averaging operator

$$\text{avg}_k(x) := \frac{1}{(3^k)^2} \sum_{a,b < 3^k} u_{a,b} x u_{a,b}^* \quad (\text{Lean: } \text{avgK})$$

is defined for every $x \in T_\infty$. On the image of level k it agrees exactly with $\tau_{\text{UHF}}(\cdot) \cdot 1$ by Theorem 5.6; combining density of the finite levels, the non-expansiveness of averaging, and the 1-Lipschitz property of τ_{UHF} yields Glimm's averaging estimate executed inside the completion:

Theorem 5.8 (`avgK_approx`). *For every $x \in T_\infty$ and $\varepsilon > 0$ there is k with $\|\text{avg}_k(x) - \tau_{\text{UHF}}(x) \cdot 1\| < \varepsilon$.*

Since two-sided ideals absorb each conjugate, avg_k preserves ideals; if an ideal I contains an element x with $\tau_{\text{UHF}}(x) \neq 0$, then Theorem 5.8 with $\varepsilon = \|\tau_{\text{UHF}}(x)\|$ combined with a Neumann-series invertibility argument forces $1 \in I$. This proves (`completion_twoSidedIdeal_bot_or_top_of_traceDetects`): if every nonzero two-sided ideal of T_∞ contains an element of nonzero trace — the hypothesis named `SubstrateCompletionTraceDetectsIdeals` — then T_∞ is simple, hence (Theorem 5.3) τ_{UHF} is faithful.

5.6. The four-fold equivalence (r108): locating the mountain. Could trace-detection be discharged directly? The formalized answer is a sharp *no*: it is not a weaker waypoint but a restatement of the summit.

Theorem 5.9 (`traceDetects_tfae`). *For the algebra T_∞ , the following are equivalent:*

- (1) trace-detection: *every two-sided ideal $I \neq \perp$ contains an x with $\tau_{\text{UHF}}(x) \neq 0$;*
- (2) *`traceNullTwoSidedIdeal` = $\perp$;*
- (3) faithfulness: *$\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$;*
- (4) simplicity: *every two-sided ideal of T_∞ is $\perp$ or $\top$.*

*The cycle is (4) $\Rightarrow$ (3) (Theorem 5.3), (3) $\Rightarrow$ (2) (`traceNull_eq_bot_of_faithful`), (2) $\Rightarrow$ (1) (`traceDetects_of_traceNull_eq_bot`), (1) $\Rightarrow$ (4) (§5.5), formalized with *mathlib*’s `List.TFAE`.*

Theorem 5.9 is the honest obstruction map: no hypothesis genuinely weaker than faithfulness itself can close the loop, and the classical hard input — Glimm’s theorem, or something of equal strength — is really required. The same development also probed the classical spectral-projection route (producing, from a nonzero selfadjoint element of an ideal, a nonzero projection in the ideal via the continuous functional calculus) and formalized exactly where it stalls in the non-unital cfc setting; the reusable API built for that probe is described in §7.3. The route that succeeded is different, and is the subject of the next section.

6. THE CONDITIONAL EXPECTATION AND THE FAITHFULNESS PROOF

The closing argument constructs the canonical conditional expectation onto each finite level, proves two contraction properties for it by elementary means, extends it to the completion, and derives faithfulness from the reconstruction property $E_k x \rightarrow x$. Simplicity then follows through Theorem 5.9. Throughout, $n = 3^k$.

6.1. The partial trace E_k (r109). Under the identification $M_{3^{n+1}}(\mathbb{C}) \cong M_{3^n}(\mathbb{C}) \otimes M_3(\mathbb{C})$ furnished by `levelStepEquiv`, the single-step partial trace $\text{id} \otimes \tau_3$ is, entrywise,

$$(\mathcal{E}B)_{ij} = \frac{1}{3} \sum_{t=0}^2 B_{(i,t),(j,t)} \quad (\text{Lean: } \text{partialTraceStep}),$$

where (i, t) abbreviates `levelStepEquiv (i, t)`. Iterating $j = k$ steps gives `partialTraceDown`: $M_{3^j} \rightarrow M_{3^k}$ for $k \leq j$, and descending through single-level representatives defines

$$E_k: \varinjlim_m A_m \longrightarrow M_{3^k}(\mathbb{C}) \quad (\text{Lean: } \text{condExp}).$$

The kernel-verified properties are the trace-preserving retraction properties needed for the faithfulness argument below; we call E_k the *level- k expectation map*. We emphasize that the full abstract conditional-expectation API — positivity, the A_k -bimodule identity $E_k(axb) = a E_k(x) b$ for $a, b \in A_k$, and idempotence as a norm-one projection — is *not* formalized here; only the properties listed in Theorem 6.1 are, and only those are used. The name “conditional expectation” below is thus informal shorthand for this partial-trace retraction, not a claim that the classical conditional-expectation axioms are machine-checked:

Theorem 6.1 (Properties of E_k). *E_k is additive and $\mathbb{C}$ -homogeneous, unital, star-preserving, inverts the embedding on the level- k corner (`condExp_level_k`: $E_k(\iota_k A) = A$, and more generally `condExp_lower_level` for levels $m \leq k$), and is trace-preserving: $\tau_{3^k}(E_k x) = \tau_{\text{pre}}(x)$ (`condExp_normalized_trace`).*

6.2. The $\|\cdot\|_2$ -contraction, elementarily (r110). The first analytic requirement is the contraction of E_k in the trace 2-seminorm:

$$(1) \quad \text{Re } \tau_{3^k}((E_k x)^*(E_k x)) \leq \text{Re } \tau_{\text{pre}}(x^*x) \quad (\text{Lean: } \text{condExp_l2_contraction}).$$

Classically (1) is the Kadison–Schwarz inequality [7] for the completely positive map E_k , composed with the trace. *Mathlib* has neither Kadison–Schwarz nor any complete-positivity

theory, and formalizing them would have been a project in itself. The formalization instead targets the *scalar* inequality directly, and it turns out to be elementary. For the single step: by finite Cauchy–Schwarz over three terms,

$$\left| \sum_{t=0}^2 a_t \right|^2 \leq 3 \sum_{t=0}^2 |a_t|^2 \quad (\text{normSq_sum_fin3_le}),$$

so each entry of $\mathcal{E}B$ obeys $|(\mathcal{E}B)_{ij}|^2 \leq \frac{1}{3} \sum_t |B_{(i,t),(j,t)}|^2$. Summing over i, j , the right-hand side is precisely the *diagonal-tail sub-sum* ($s = t$) of

$$\tau_{3^{n+1}}(B^*B) = \frac{1}{3^{n+1}} \sum_{i,s,j,t} |B_{(j,t),(i,s)}|^2,$$

all of whose terms are nonnegative; `mathlib`’s `Finset.single_le_sum` (in its sub-sum form) closes the single-step inequality $\tau_{3^n}((\mathcal{E}B)^*(\mathcal{E}B)) \leq \tau_{3^{n+1}}(B^*B)$ (`partialTraceStep_normSq_trace_le`). Iterating through `partialTraceDown` and descending to the direct limit gives (1). No positive-semidefinite operator theory and no complete positivity enter at any point — a useful proof-engineering simplification: for trace-compatible partial traces, the scalar 2-norm contraction is a three-line Cauchy–Schwarz fact, so the Kadison–Schwarz detour that the textbook proof takes is avoidable in the formalization.

6.3. The operator-norm contraction via isometries (r111). Extending E_k to the completion requires uniform continuity, i.e. the operator-norm bound $\|E_k x\| \leq \|x\|$. Classically this is “a unital completely positive map has norm one,” again unavailable. The formalized proof uses the isometry (Stinespring-flavoured [12]) decomposition of the partial trace. For $t \in \{0, 1, 2\}$ let $W_t: \mathbb{C}^{3^n} \rightarrow \mathbb{C}^{3^{n+1}}$ be the coordinate injection $v \mapsto v \otimes e_t$, as a $3^{n+1} \times 3^n$ matrix (Lean: `Winj`). Then

$$(2) \quad \mathcal{E}B = \frac{1}{3} \sum_{t=0}^2 W_t^* B W_t \quad (\text{partialTraceStep_eq_smul_sum}),$$

and each W_t satisfies $W_t^* W_t = I$ (`Winj_conjTranspose_mul_self`). The C^* -identity for the L^2 -operator norm gives $\|W_t\|^2 = \|W_t^* W_t\| = \|I\| \leq 1$ (`Winj_opNorm_sq_le_one`, using `Matrix.l2_opNorm_conjTranspose_mul_self` and `ContinuousLinearMap.norm_id_le`), so submultiplicativity and $\|W_t^*\| = \|W_t\|$ yield $\|W_t^* B W_t\| \leq \|B\|$, and averaging over the three slots closes

$$\|\mathcal{E}B\| \leq \|B\| \quad (\text{partialTraceStep_opNorm_le}),$$

hence, iterating, $\|E_k x\| \leq \|x\|$ (`condExp_opNorm_le`). Together with additivity this makes E_k 1-Lipschitz (`condExp_lipschitz`), hence uniformly continuous. The only `mathlib` inputs are four L^2 -operator-norm lemmas and the norm bound on the identity operator; no completely-positive-map theory is used.

6.4. Extension to the completion and the closing argument (r112). Uniform continuity extends E_k to the completion in two guises. First, the matrix-valued extension

$$\text{condExpFin } k: T_\infty \rightarrow M_{3^k}(\mathbb{C}), \quad \text{condExpFin } k := \text{UniformSpace.Completion.extension}(E_k).$$

Second, its recoordinatization back into the completion,

$$\text{condExpCompletion } k: T_\infty \rightarrow T_\infty, \quad x \mapsto \iota_k(\text{condExpFin } k \, x).$$

Both are 1-Lipschitz (`condExpFin_lipschitz`, `condExpCompletion_lipschitz`, the latter through the isometry of ι_k , Theorem in §3), and agreeing with E_k on the dense image (`condExpFin_coe`, `condExpCompletion_coe`). Three statements assemble the proof.

Lemma 6.2 (Reconstruction: `condExpCompletion_tendsto`). *For every $x \in T_\infty$, $\text{condExpCompletion } k \, x \rightarrow x$ as $k \rightarrow \infty$.*

Proof sketch. An $\varepsilon/2$ density lift over the uniformly 1-Lipschitz family: approximate x within $\varepsilon/2$ by a dense-image element with single-level representative at level m ; for $k \geq m$ the map `condExpCompletion k fixes` that element (`condExpCompletion_fixes_coe_lower`, from `condExp_lower_level` and the cocone identity), and the Lipschitz bound controls the difference at x . $\square$

Lemma 6.3 (The lifted $\|\cdot\|_2$ -contraction: `condExpCompletion_l2_contraction`). *For every k and $x \in T_\infty$,*

$$\text{Re } \tau_{\text{UHF}}((\text{condExpCompletion } k \ x)^*(\text{condExpCompletion } k \ x)) \leq \text{Re } \tau_{\text{UHF}}(x^*x).$$

Proof sketch. The inequality between two continuous real-valued functions of x holds on the dense image by (1) and the trace bridge $\tau_{\text{UHF}}((E_k x)^*(E_k x)) = \tau_{3^k}((\text{condExpFin } k \ x)^*(\text{condExpFin } k \ x))$ (`UHF_trace_condExpCompletion`, definitional since `condExpCompletion` factors through ι_k); completion induction on the closed set $\{x : f(x) \leq g(x)\}$ (`isClosed_le`) transfers it to all of T_∞ . $\square$

Lemma 6.4 (Matrix faithfulness in real-part form: `matrix_faithful_of_re_nonpos`). *If $\text{Re } \tau_n(M^*M) \leq 0$ then $M = 0$.*

Proof sketch. Positivity pins $\tau_n(M^*M)$ to the nonnegative reals; a nonpositive real part then forces $\tau_n(M^*M) = 0$, and matrix-tier faithfulness ($\Phi 1$) concludes. $\square$

Theorem 6.5 (The summit: `UHF_trace_faithful`). *For every $x \in T_\infty$: if $\tau_{\text{UHF}}(x^*x) = 0$ then $x = 0$. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`; zero project axioms.*

Proof. Suppose $\tau_{\text{UHF}}(x^*x) = 0$. By Lemma 6.3 and the trace bridge, $\text{Re } \tau_{3^k}((\text{condExpFin } k \ x)^*(\text{condExpFin } k \ x)) \leq 0$ for every k ; by Lemma 6.4, `condExpFin k x = 0`, hence `condExpCompletion k x = 0`, for every k . By Lemma 6.2 the constant-zero sequence `condExpCompletion k x` converges to x ; uniqueness of limits (`tendsto_nhds_unique`) gives $x = 0$. $\square$

Corollary 6.6 (Glimm simplicity, machine-checked: `substrate_completion_simple_unconditional`). *Every two-sided ideal of T_∞ is $\perp$ or $\top$.*

Proof. Theorem 6.5 is item (3) of the four-fold equivalence (Theorem 5.9); item (4) follows. In Lean the step is literally `traceDetects_iff_simple.mp` $\circ$ `traceDetects_iff_faithful.mpr` applied to `UHF_trace_faithful`. $\square$

Remark 6.7. The classical order of deduction is inverted: Glimm proves simplicity first and derives faithfulness of the trace; here faithfulness is proved directly through the conditional expectations, and simplicity follows. Both directions of the equivalence are formalized, so the classical order is also machine-checked — Theorem 5.3 is exactly Glimm’s implication. Note also that Corollary 6.6 is *algebraic* simplicity (all two-sided ideals, not only closed ones), the stronger classical statement, which holds for all UHF algebras [4].

Remark 6.8 (Residuals discharged). Theorem 6.5 discharges, as kernel theorems, every named hypothesis left standing earlier in the arc: the conditional-expectation extension package (`condExpCompletionExtension_holds`), the r100 completion-faithfulness residual (`substrate_UHF_completion_positive_faithfulness`), and trace-detection (`substrate_completion_traceDetectsIdeals`). The capstone theorem `r112_substrate_completion_faithful_capstone` bundles the seven statements — extension, reconstruction, lifted contraction, faithfulness, the two residuals, and simplicity — as a single verified conjunction.

6.5. Uniqueness of the trace (r113). The same conditional-expectation machinery yields the third leg of the classical UHF identification: τ_{UHF} is the *only* tracial state on T_∞ . We package the hypotheses as a predicate `IsTracialState` φ requiring φ to be continuous, additive, $\mathbb{C}$ -homogeneous, unital ($\varphi(1) = 1$), and tracial ($\varphi(xy) = \varphi(yx)$); positivity is deliberately omitted.

Theorem 6.9 (Uniqueness of the substrate trace). *For every φ with `IsTracialState` φ and every $x \in T_\infty$, $\varphi(x) = \tau_{\text{UHF}}(x)$ (`substrate_UHF_trace_unique`). Kernel axioms [`propext`, `Classical.choice`, `Quot.sound`].*

The proof is the standard argument, formalized in three steps. *Finite level:* any additive, $\mathbb{C}$ -homogeneous, unital, tracial functional on $M_n(\mathbb{C})$ equals the normalized trace (`matrix_tracial_state_unique`) — the off-diagonal matrix units are commutators $E_{ij} = [E_{ik}, E_{kj}]$ and hence annihilated, while the diagonal units are cyclic-conjugate and hence equal, so φ is pinned to $\varphi(1)/n$ times the trace; unitality fixes the constant. Mathlib does not carry this lemma, so it is proved directly from `Matrix.single` identities and `matrix_eq_sum_single`. *Level lift:* for $x = \iota_k(A)$, $\varphi \circ \iota_k$ is such a functional, so $\varphi(x) = \tau_{3^k}(A) = \tau_{\text{UHF}}(x)$. *Density lift:* φ and τ_{UHF} are continuous and agree on the dense direct-limit image, so they agree everywhere (`isClosed_eq` on the completion, exactly as in §6.4).

Combining Theorem 6.5, the simplicity corollary, and Theorem 6.9, the capstone `r113_substrate_UHF_factor_capstone` records that T_∞ is a faithful, simple, uniquely-traced UHF C^* -algebra — i.e. the Glimm 3^∞ UHF factor.

7. FORMALIZATION NOTES

7.1. Mathlib infrastructure used. The construction is deliberately mathlib-native; the load-bearing APIs were:

- `Ring.DirectLimit` with `DirectLimit.lift` and `DirectLimit.exists_eq_mk` (single-level representatives, used in essentially every direct-limit argument);
- `UniformSpace.Completion`: `extension`, `extension_coe`, `induction_on` and `induction_on_on2` (closed-set completion induction), `denseRange_coe`, `coe_isometry`;
- the scoped L^2 -operator-norm C^* -structure on matrices (`Matrix.Norms.L2Operator`): `l2_opNorm_mul`, `l2_opNorm_conjTranspose`, `l2_opNorm_conjTranspose_mul_self`, `cs_tar_norm_def`;
- matrix trace and positivity lemmas: `Matrix.trace_kronecker`, `trace_mul_comm`, `trace_conjTranspose`, `trace_diagonal`, `trace_conjTranspose_mul_self_eq_zero_iff`, `posSemidef_conjTranspose_mul_self`, `PosSemidef.trace_nonneg`;
- `IsSimpleRing.matrix` and `TwoSidedIdeal` (lattice structure, `mk'`, `eq_bot_or_eq_top`);
- roots of unity (`Complex.isPrimitiveRoot_exp`, `IsPrimitiveRoot.pow_inj`, `geom_sum_mul`) for the Weyl engine;
- the scoped `ComplexOrder` with `Complex.orderClosedTopology` (positivity as a closed condition), and `discrim_le_zero` (Cauchy–Schwarz);
- for the API files of §7.3: the non-unital continuous functional calculus (`cfc_n`, `cfc_nHom`), `CStarAlgebra.approximateUnit`, and the Stone–Weierstrass induction principle `ContinuousMapZero.induction_on_of_compact`.

7.2. Gaps found in mathlib. Three absences shaped the mathematics of §§5–6 and are, we suggest, the natural next targets for mathlib’s operator-algebra development:

- (G1) **No completely positive map theory.** Mathlib has no notion of complete positivity, no Kadison–Schwarz inequality, no partial trace, and no “unital CP map has norm one.” This forced — and, in the end, rewarded — the elementary routes of §6.2 (three-term Cauchy–Schwarz instead of Kadison–Schwarz) and §6.3 (isometry decomposition instead of CP norm theory). The two proofs are shorter than the classical ones and need a fraction of the theory; they may be of expository interest independent of the formalization.
- (G2) **No spectral projections.** The Borel functional calculus and spectral projections are absent; the continuous functional calculus exists but produces projections only from *clopen* spectral sets. The Route-A probe of §5.6 built exactly that (§7.3) and identified finite-quasispectrum witnesses as the missing hypothesis — a wall that the conditional-expectation route then bypassed entirely.

(G3) No C^* -order on completions. T_∞ , built as a `UniformSpace.Completion`, does not automatically carry the `PartialOrder/StarOrderedRing` structure of a C^* -algebra order; positivity of τ_{UHF} was therefore formulated and proved scalar-side, in the `ComplexOrder` on $\mathbb{C}$, rather than operator-side. An order-instance transfer for completions of normed $*$ -algebras would let statements like Theorem 4.1(v) take their textbook form.

7.3. Contributed API: the `ForMathlib` files. Three standalone files, written to mathlib contribution standards (mathlib copyright header, full docstrings, no project dependencies beyond one another), were produced during the Route-A probe and are offered upstream:

- `PF/ForMathlib/TwoSidedIdealClosure.lean` — `TwoSidedIdeal.closure`: the topological closure of a two-sided ideal in a (non-unital, non-associative) topological ring is a two-sided ideal, with the expected API (`le_closure`, `closure_minimal`, `closure_eq_of_isClosed`) and, in a unital topological ring with open unit group (e.g. a Banach algebra), properness of the closure of a proper ideal (`closure_ne_top`). Mirrors mathlib’s `Ideal.closure` for one-sided ideals.
- `PF/ForMathlib/CfcMemTwoSidedIdeal.lean` — the non-unital continuous functional calculus maps into closed two-sided ideals: for a selfadjoint in a closed two-sided ideal I of a non-unital C^* -algebra and *any* $f: \mathbb{R} \rightarrow \mathbb{R}$, $\text{cfc}_n f a \in I$ (`TwoSidedIdeal.cfc_c_n_mem_of_isClosed`), unconditional thanks to the junk-value convention. The proof runs an approximate-unit argument for scalar closure (`smul_mem_of_isClosed`) and Stone–Weierstrass induction over $C(\sigma_n \mathbb{R} a, \mathbb{R})_0$.
- `PF/ForMathlib/ClopenSpectralProjection.lean` — `spectralProjection a U := cfc_n(U.indicator 1) a` for U clopen in the quasispectrum with $0 \notin U$: selfadjoint, idempotent, nonzero iff U meets the quasispectrum, and a member of every closed two-sided ideal containing a — i.e., clopen spectral data away from 0 manufactures a genuine projection inside a nonzero ideal.

7.4. Companion result. The same repository formalizes, independently of everything above, a sesquilinear self-adjointness identity for a symmetrised base-3 transfer operator on the mathlib Hilbert space $L^2([0, 1], dx/x)$ (theorem `T3_self_adjoint_conj`; see the companion ITP note and [3]). It shares the base-3 architecture but is logically disjoint from the UHF development: neither result uses the other. We record it here only for completeness of the repository’s verified operator-algebra content.

8. REPRODUCIBILITY

Pinned toolchain.

Lean toolchain: `leanprover/lean4:v4.24.0-rc1`

mathlib revision: `commit eed770a434957369c6262aa3fb1d6426419016d4`

Repository: `github.com/FractalDevTeam/Principia-Fractal`, branch `master`

HEAD reference (verified): `91cc1a01 (2026-07-23)`; build and `#print axioms` run same-day Build commands.

```
git clone https://github.com/FractalDevTeam/Principia-Fractal.git
cd Principia-Fractal/PF_Lean4_Code
elan install $(cat lean-toolchain)
lake build PF      # 4,472 jobs
lake build         # 8,930 jobs (adds PF_Lean4Lean re-elaboration)
```

Headline theorems and `#print axioms` output. The axiom audit blocks at the end of `PF/SubstrateCompletionFaithful.lean` (and the corresponding blocks in the `r102–r111` files) report, for every theorem cited in this paper:

```
'PrincipiaTractalis.SubstrateCompletionFaithful.UHF_trace_faithful'
  depends on axioms: [propext, Classical.choice, Quot.sound]
```

```
'PrincipiaTractalis.SubstrateCompletionFaithful.
```

```
substrate_completion_simple_unconditional'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

```
'PrincipiaTractalis.SubstrateCompletionFaithful.
r112_substrate_completion_faithful_capstone'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

These are the three standard Lean 4 kernel axioms (propositional extensionality, choice, quotient soundness); there are no project-level axioms, and zero `sorry` occurrences on any code path leading to the theorems above. The `PF_Lean4Lean` package (separate `lakefile.toml` and package hash, same pinned `mathlib`) re-elaborates each theorem through the production Lean 4 kernel under a distinct package boundary, adding roughly 4,400 further verification jobs.

Where to find each result. Table 1 maps the paper’s sections to source files.

Content	Paper	Lean file (under PF/)
Tower, limit, completion, instances	§3	<code>Substrate*</code> (r41–r60 chain)
Tracial state and soft properties	§4	<code>SubstrateUHFTraceIs*.lean</code>
Cauchy–Schwarz for the trace	§5.1	<code>SubstrateUHFTraceCauchySchwarz.lean</code>
Trace-null ideal; faithful-of-simple	§5.2	<code>SubstrateUHFTraceNullIdeal.lean</code>
Weyl averaging engine	§5.3	<code>SubstrateMatrixUnitaryAveraging.lean</code>
Direct-limit simplicity	§5.4	<code>SubstrateDirectLimitSimplicity.lean</code>
Completion averaging	§5.5	<code>SubstrateCompletionSimplicity.lean</code>
Four-fold equivalence	§5.6	<code>SubstrateTraceDetectionAttempt.lean</code>
Conditional expectation E_k	§6.1	<code>SubstrateConditionalExpectation.lean</code>
$\ \cdot\ _2$ -contraction	§6.2	<code>SubstrateCondExpContraction.lean</code>
Operator-norm contraction	§6.3	<code>SubstrateCondExpOpNorm.lean</code>
Extension; faithfulness; simplicity	§6.4	<code>SubstrateCompletionFaithful.lean</code>
Contributed API	§7.3	<code>ForMathlib/*.lean</code>

TABLE 1. Section-to-source map.

9. RELATED AND FUTURE WORK

9.1. Related work. `Mathlib`’s C^* -algebra hierarchy, continuous functional calculus, matrix positivity infrastructure, direct limits, and completions — all community developments, with the functional-analysis foundations described in [5] — are the platform this work builds on and would have been impossible without. To the best of our knowledge, however — based on keyword and definition searches of the `mathlib` library, the Coq `opam` package archive, and the Isabelle Archive of Formal Proofs for UHF/AF/Glimm/simple- C^* /tracial-state content, and on the operator-algebra formalization literature — no prior formalization in Lean, Coq, or Isabelle contains a concrete infinite-dimensional simple C^* -algebra, a canonical faithful tracial state on one, or any version of Glimm’s simplicity theorem [6]. This does not constitute an exhaustive survey of every ITP library, and we would welcome correction from operator-algebra formalization specialists. On the classical side, everything proved here is textbook material [4, 1, 13]: Glimm’s classification and simplicity [6], Powers’ work on the associated factors [11], Bratteli’s AF framework [2], and the conditional-expectation theory of Tomiyama [14] descending from Kadison [7] and Stinespring [12]. The contribution is not new mathematics about $M_{3\infty}$; it is the first machine-checked path to these facts, together with the elementary replacement proofs (§6.2, §6.3) that the machine-checked setting demanded and produced.

9.2. Future work. Natural next steps, roughly in order of proximity:

- *Upstreaming*: the three `ForMathlib` files (§7.3), and de-specialising the tower construction from base 3 to an arbitrary supernatural number — most proofs are already base-generic in structure.

- *Uniqueness of the trace*: τ_{UHF} is canonical by construction; formalizing that it is the *only* tracial state on T_∞ (classically via the averaging engine of §5.3, which is already formalized) would complete the textbook triple simple/unique-trace/faithful.
- *CP-map foundations*: gap (G1) — partial traces, complete positivity, Kadison–Schwarz — now has a worked concrete example to test an abstract API against.
- *GNS and the hyperfinite II_1 factor*: the GNS representation of τ_{UHF} , the $\|\cdot\|_2$ -completion, and the trace on the double commutant, connecting to the Murray–von Neumann hyperfinite factor [10, 11].
- *Glimm’s classification*: isomorphism invariance via the supernatural number, for which the clopen spectral projection API of §7.3 is a first ingredient.

ACKNOWLEDGMENTS

The Lean 4 and mathlib community built the C^* -algebra, direct-limit, completion, and functional-calculus infrastructure on which every theorem in this paper stands.

REFERENCES

- [1] B. Blackadar. *Operator Algebras: Theory of C^* -Algebras and von Neumann Algebras*. Encyclopaedia of Mathematical Sciences 122, Springer, 2006.
- [2] O. Bratteli. Inductive limits of finite dimensional C^* -algebras. *Trans. Amer. Math. Soc.*, 171:195–234, 1972.
- [3] P. Cohen. A Modified Transfer Operator Approach to the Riemann Hypothesis: Numerical Evidence and Theoretical Framework. Manuscript, June 12, 2025. Preserved at [Papers/PriorWork_Cohen2025_TransferOperatorRH/](#) in the source repository.
- [4] K. R. Davidson. *C^* -Algebras by Example*. Fields Institute Monographs 6, American Mathematical Society, 1996.
- [5] F. Dupuis, R. Y. Lewis, and H. Macbeth. Formalized functional analysis with semilinear maps. In *13th International Conference on Interactive Theorem Proving (ITP 2022)*, LIPIcs 237, 2022.
- [6] J. Glimm. On a certain class of operator algebras. *Trans. Amer. Math. Soc.*, 95:318–340, 1960.
- [7] R. V. Kadison. A generalized Schwarz inequality and algebraic invariants for operator algebras. *Ann. of Math. (2)*, 56:494–503, 1952.
- [8] The mathlib Community. The Lean Mathematical Library. In *Proc. CPP 2020*, pages 367–381, 2020.
- [9] L. de Moura and S. Ullrich. The Lean 4 Theorem Prover and Programming Language. In *CADE 28*, 2021.
- [10] F. J. Murray and J. von Neumann. On rings of operators. *Ann. of Math. (2)*, 37(1):116–229, 1936.
- [11] R. T. Powers. Representations of uniformly hyperfinite algebras and their associated von Neumann rings. *Ann. of Math. (2)*, 86:138–171, 1967.
- [12] W. F. Stinespring. Positive functions on C^* -algebras. *Proc. Amer. Math. Soc.*, 6:211–216, 1955.
- [13] M. Takesaki. *Theory of Operator Algebras I*. Springer, 1979.
- [14] J. Tomiyama. On the projection of norm one in W^* -algebras. *Proc. Japan Acad.*, 33:608–612, 1957.
- [15] H. Weyl. Quantenmechanik und Gruppentheorie. *Zeitschrift für Physik*, 46:1–46, 1927.

MESA, ARIZONA, USA

Email address: psolorzano@alumni.berklee.edu

URL: [ORCID:~0009-0002-0734-5565](https://orcid.org/0009-0002-0734-5565)